I realize that the discussion, so far, it's been quite abstract; to understand what's the real purpose of computing these values for the reachability distance and the ordering of processing for the dataset, and how they can be used to build a hierarchical clustering, we need to get ahead of ourselves and take a look at figure 12.17, that shows a *reachability plot*, that can be considered the output of OPTICS (and also requires a further step after the main algorithm, as we'll see in next section).

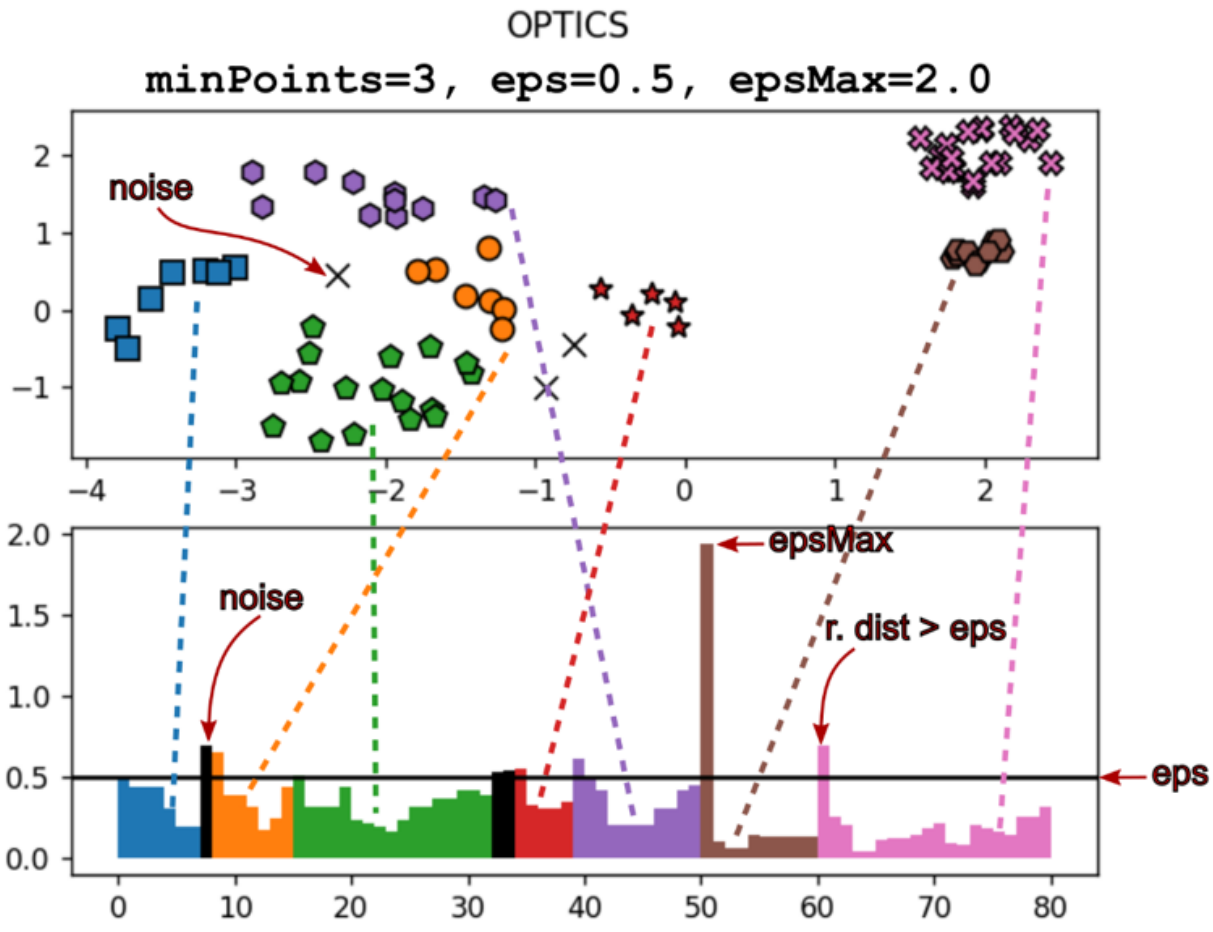


Figure 12.17 A reachability plot. The top chart shows the clustered dataset, the bottom half the reachability distances of points in the dataset, in the same order as points are processed by OPTICS.